

Name	Jenica A	Register No.	192425003
Course	ITA0510 - COMPUTER VISION	Experiment No.	11
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Institution	Saveetha School of Engineering, SIMATS	Signature	

EXPERIMENT 11

Affine Transformation on an Image

AIM

To perform an affine transformation on the given image.

PROGRAM

```
import cv2
import numpy as np

img = cv2.imread("image1.jpg")
rows, cols = img.shape[:2]

# An affine transform is defined by three corresponding points
src_points = np.float32([[50, 50], [200, 50], [50, 200]])
dst_points = np.float32([[10, 100], [200, 50], [100, 250]])

M = cv2.getAffineTransform(src_points, dst_points)
affine_img = cv2.warpAffine(img, M, (cols, rows))

cv2.imwrite("affine_transformed.jpg", affine_img)
print("Affine matrix:\n", M)

cv2.imshow("Original", img)
cv2.imshow("Affine Transformed", affine_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT



image1.jpg (input image)

OUTPUT



affine_transformed.jpg (affine transform: parallel lines stay parallel)

RESULT

Thus an affine transformation was successfully applied to the given image using `cv2.getAffineTransform()` and `cv2.warpAffine()`.